

Treasury auction method and underpricing: Evidence from Iceland

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Abstract

The Central Bank of Iceland replaced the discriminatory method used for auctioning treasury securities with the uniform-price method in 2009. We analyze underpricing before and after this institutional reform with a sample of 516 auctions organized from 2000 to 2018. After controlling for auction characteristics and financial market conditions, we find that underpricing is lower under the uniform-price method. However, this underpricing decline does not translate into a reduction in sovereign issuance cost. The emergence of overpricing, observed in the late part of our sample, coincides with the growing importance of commissions paid to primary dealers. Our results provide practical implications for governments, regulators, and market participants.

JEL CLASSIFICATION

G14, G18, H63

1 | INTRODUCTION

Governments issue thousands of treasury bills and bonds every month to finance sovereign debt. Primary dealers, who participate in the primary market (the auction), usually hold a fraction of the treasury securities awarded and resell the rest in the secondary market to various investors. A positive difference between the price in the secondary and primary markets can be viewed as a shortfall for the auctioneer, who stands to benefit more by issuing directly to secondary market participants. Hence, such mispricing involves a wealth transfer from issuers to primary dealers. In such a setting, ideas for reducing the shortfall might draw the attention of governments and regulators. In Friedman (1960), Milton argued that the choice of the auction method, by influencing the bidding strategy of each player and the likelihood of collusion between players, determines the auction outcome. Ultimately, the mispricing level is impacted.

Our paper links the choice of the auction method to the magnitude of mispricing. In 2008, the Central Bank of Iceland modified the auction method for the issuance of Government of Iceland bills and bonds. The discriminatory method was replaced with a new uniform-price method.¹ Our paper uses this institutional change in order to assess which method is better when the auctioneer seeks to minimize the cost of sovereign debt issuance. If the introduction of the uniform-price method lowers this cost significantly, after controlling for various other factors, it provides evidence of the superiority of the uniform-price method over the discriminatory one. By analyzing an exclusive dataset of Icelandic sovereign bond and bill auctions around the switch of auction method, we provide evidence that the uniform-price method is better for the issuer. Under the discriminatory method, the average underpricing is 7.5 cents for 100 Icelandic Krona of face value. Under the uniform-price one, underpricing is -0.5 cents, on average.

Ausubel, Cramton, Pycia, Rostek, and Weretka (2014) state that the ranking of the two auction methods is ambiguous, whether using efficiency or revenue maximization as a criterion. Bidder numbers, risk aversion, and informational asymmetry drive the ranking jointly. It follows that determining the superiority of one method over another depends on the particular features of each country's treasury auctions, and thus should be addressed empirically. We follow the suggestion of Ausubel, et al. (2014) and directly assess the strength of the statistical association between auction performance and auction method, using a panel of 516 auctions and controlling for time-varying changes in exogenous factors. We find that the uniform-price method is significantly associated with less revenue transfer from the issuer to the bidders.

Our data is very unique in four ways. First, our primary market data covers a long period of time, almost two decades of auctions. Second, the data contains bid-level information. We observe, by bidder, winning bids as well as losing ones, quantities demanded and awarded. Third, each primary dealer is assigned a unique ID consistent across auctions and over time. Fourth, Nasdaq Nordic provides us with high-quality secondary market data, including transaction prices, which makes our measures of underpricing very precise. Transaction prices are rare to have as many studies suffer from using indicative bid, ask, or mid prices. Since some transactions take place inside or outside the bid-ask spread, it casts doubt on the reliability of underpricing generated with secondary market quotes.

An interesting feature of the Icelandic Treasury market is the low number of primary dealers. The number of auction participants is large in the empirical studies mentioned above, and to the best of our knowledge, there is no evidence of underpricing for treasury auctions characterized by either the absence of foreign primary dealers or the low number of bidders resulting from this absence. Most empirical studies on the treasury auction method analyze the US treasury market, which is characterized by a very large number of primary dealers participating in auctions (24 in 2021). The fact is that most of the countries in the world have fewer participants. By gathering information on 49 governmental agencies and central banks, we find that the average number of primary dealers is 12, and 30 percent of the countries have less than 10 primary dealers. It turns out that only eight financial institutions have ever been accredited as primary dealers by the Central Bank of Iceland between 2000 – 2018, and the median number of primary dealers participating in an auction is only four. Hence, our study is relevant and useful for many countries, especially for emerging markets.²

To shed light on the reduction of underpricing following the introduction of the uniform-price method, we perform several non-parametric tests and run a series of multivariate regressions. We control for the explanatory variables commonly used in the underpricing literature: auction characteristics, treasury security characteristics, and market conditions. We also improve the model specification by controlling for time-varying changes in commissions

¹These are the most common methods used by governments or central banks worldwide for sovereign bond issuance. Under the discriminatory method, also called the multiple-price method, securities are awarded to the highest bidder, and the allocations continue along the demand curve up to the amount offered is fully covered. The winning bidders pay what they bid. Under the uniform-price method, also called the single-price method, all winning bidders pay the same price, which is the lowest bid exhausting the supply. This price is called the stop-out price or the market-clearing price. Interchangeably, discriminatory auctions are also referred to as multiple-price auctions and uniform-price auctions are also referred to as single-price auctions in the literature.

²The following countries have less than 6 primary dealers: Botswana, Bulgaria, Jamaica, Georgia, Mauritius, South Africa, and Uruguay.

paid to primary dealers and the extent of the security lending program. To the best of our knowledge, the level of commission has never been used as an explanatory variable, which is surprising as numerous auctioneers in the world pay commissions to auction participants.³ Likewise, while many treasury auctioneers have developed security lending programs, published studies ignore the influence of the amount loaned in each treasury security on that security's underpricing. Our results show that these two variables mobilize predictive powers to explain underpricing. This new finding deserves attention as it suggests that the auction method is not the only institutional setting the auctioneer can change to influence the issuance cost. An appropriate compensation package for primary dealers and a generous lending program might help reduce the issuance cost.

The literature shows no single way of calculating underpricing. In order to persuade that our main result holds, regardless of the way underpricing is calculated, we perform two robustness checks. First, we calculate this variable by replacing prices with implied yield to maturity. Second, we take into account the possibility that the difference between the auction outcome price and the secondary market price observed post-auction might be induced by new information. We derive an adjusted measure of underpricing by removing the expected informational component, following Hamao and Jegadeesh (2002). Our results show that underpricing is significantly lower under the uniform-price method regardless of how underpricing is measured.

In the last section, we discuss the mechanisms that might explain the association between the uniform-price method and less underpricing. Since underpricing is computed with two prices, the primary and secondary market prices, we also analyze whether secondary market price patterns are affected by the auction method change. We find that the auction method influences bid shading magnitude. It seems that the (primary dealers) PDs are more (less) aggressive under the uniform-price (discriminatory) method, and post more (less) competitive bids. This result is consistent with the winner's curse hypothesis (Bikhchandani and Huang, 1993; Milgrom, 1989). We also find that the issuance of a treasury security generates a significant price impact on the secondary market, but under the discriminatory method only. Therefore, increased bid shading by prudent PDs within the primary market, in conjunction with heightened price impact in the secondary market, correlates with a greater propensity for the discriminatory pricing method to generate higher levels of underpricing.

2 | TREASURY AUCTION METHODS: A LITERATURE REVIEW AND CURRENT PRACTISES

A long-standing debate in the treasury auction literature focuses on the ranking of auction methods. One reason is that this issue closely links to studies of single-unit auctions, an area where abundant research has addressed the ranking of different auction methods (Vickrey, 1961). Treasury auctions belong to the more general category of multi-unit auctions in which more than one identical unit of goods (bonds and bills in this case) are sold. The other, more important reason is that the monetary amounts involved in treasury auctions are so large that a tiny improvement in auction performance would generate large amounts of additional revenue for central governments, therefore lowering the cost of debt financing. Within the multi-unit auction setting, governments choose between two widely used auction methods to issue debt: the discriminatory method and the uniform-price method. Roughly speaking, these methods correspond to the first-price and second-price methods in single-unit auctions, respectively. In a discriminatory auction, winning bidders pay their own bids individually, whereas in a uniform-price auction, all winning bidders pay the lowest winning price, called the stop-out price.

Theoretical research on treasury auctions still does not provide conclusive revenue rankings between these two methods. See, among others, Wilson (1979), Milgrom and Weber (1982), Bikhchandani and Huang (1989, 1993), Back and Zender (1993), Chatterjea and Jarrow (1998), Wang and Zender (2002), Kremer and Nyborg (2004), Nyborg and

³Eisl, Ochs, Staghøj, and Subrahmanyam (2022) show the introduction of a compensation package offered to liquidity providers in the Danish sovereign bond market in 2017 has impacted liquidity premia and bond yields, suggesting that a remuneration plan might influence the behavior of a primary dealer.

Strebulaev (2004), Kang and Puller (2008), and Kastl (2011). In particular, Back and Zender (1993) compare the two methods when collusive behavior is possible. They show that implicit collusion among bidders is likely to be more difficult in an auction under the discriminatory method, which may result in higher expected revenue compared to that generated in a uniform-price auction. Nyborg and Strebulaev (2004) compare the two methods in a setting where bidders can be short-squeezed⁴ in the secondary market. They show that the likelihood of being short-squeezed is higher under the discriminatory method, which fosters bidding aggressiveness in the auction, producing larger issuer revenues. Goswami, Noe, and Rebello (1996), Sade, Schnitzlein, and Zender (2006), and Morales-Camargo, Sade, Schnitzlein, and Zender (2013) organize experimental sessions with students and professionals in order to compare auction performance under the two methods. Their conclusion differs: In Sade, et al. (2006), the discriminatory method generates the lowest average revenue while in Goswami, et al. (1996) and in Morales-Camargo, et al. (2013), it outperforms the uniform-price method.

Empirically, the ranking of auction methods is based on auction performances, with a particular focus on underpricing. Underpricing in treasury auctions, similar to equity in initial public offerings (IPOs), is computed as the percentage difference in price between what winning bidders pay in auction for one security and what they earn by selling this security in the secondary or when-issued market. Underpricing is also called the price premium, the markup, or the margin. It can be viewed as a shortfall for the treasury security issuer, as it is the extra revenue he would have earned had the securities been sold directly in the secondary market. In the context of sovereign bond issuance, positive underpricing implies revenue transfer from national governments to PDs who participate in the primary market and act as market-makers in the secondary market. If we can identify which factors impact the underpricing, we can offer recommendations for auction organizers to reduce the revenue transfer.

The US Treasury switched from the discriminatory method to the uniform-price method for certain maturities during the 1990s, and it completely switched the method for all its auctions in October 1998. These two events offer an interesting laboratory to assess the method ranking empirically and several papers analyze underpricing around these events. Interestingly, their conclusion diverges. On one side, Simon (1994), focusing on the historical mid-1970s method change of the US Treasury auctions, finds evidence in favor of the discriminatory method. On the other side, Nyborg and Sundaresan (1996) and Archibald and Malvey (1998), studying the second US method change in 1992, advocate for the replacement of the discriminatory method with that of uniform-price. Goldreich (2007), who leads his analysis over a large time window, spanning the 1992 experimentation and the 1998 complete switch, finds that the uniform-price method produces lower issuance costs for the US government. Apart from the US Treasury case, there are few published academic studies on the subject of auction method switching. Umlauf (1993) analyses underpricing (called profit margin in his paper) of a series of auctions that took place around July 1990, when the Mexican Treasury replaced the discriminatory method with the uniform-price one. He finds that the uniform, in contrast to discriminatory, auctions appear to raise more revenue. Kang and Puller (2008) study a series of auctions around a similar switch made by the Korean Treasury in August 2002. They find that discriminatory auctions produce higher expected revenue and better security allocation.

To overcome the lack of events on the treasury auction method change, some papers have developed auction models that make counterfactual exercises possible, and quantify what would have been the benefit/loss for the auction participants if the uniform-price method replaced the discriminatory one. Interestingly, their conclusion diverges. Armantier and Sbaï (2006) and Heller and Lengwiler (2001) show that shifting from the discriminatory to the uniform-price method would reduce the government financing cost, whereas Hortaçsu and McAdams (2010) and Marszalec (2017) provide evidence in favor of the superiority of the discriminatory method. Hence, empirically, neither counterfactual analysis nor underpricing analysis provides a clear answer to which auction method is the best. How much governments could benefit or lose following a change in auction method is still an open question with no consensus.

⁴A short-squeeze happens when a PD obtains a significant fraction of the issued securities and set the price abusively high when trading against PDs with large pre-auction short positions.

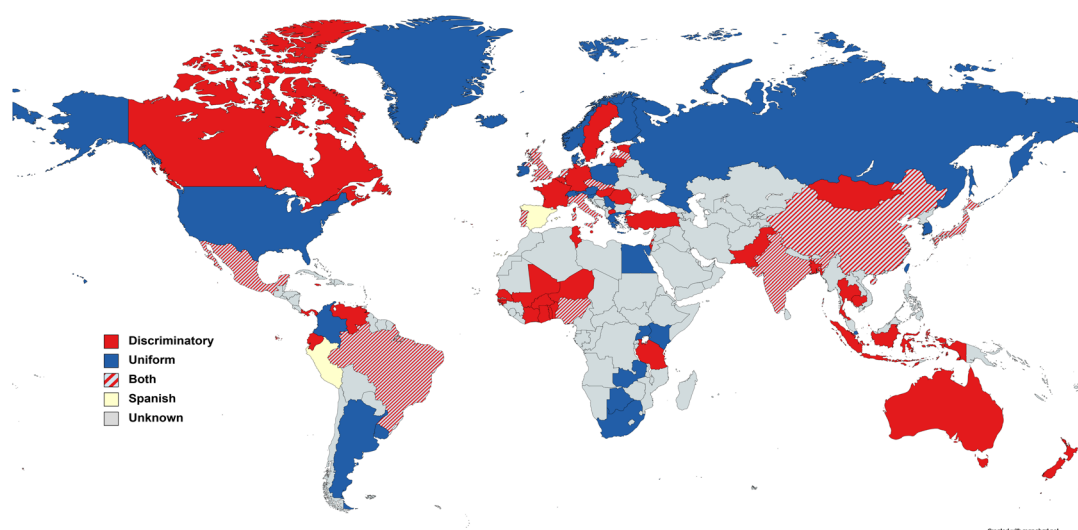


FIGURE 1 Use of the Treasury Auction Method in the World. This map shows the choice of the treasury auction method. Data was collected from governmental agencies and central banks in November 2021. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jfr.12428)]

Verifying the current practices is helpful to see whether one auction method is more popular than the other worldwide. We collect information across government agencies and central banks in charge of the issuance process. Figure 1 shows a world map of the auction security method used across 59 countries for 2021. It shows that practice is divided, with 46% using the discriminatory method and 39% using the uniform-price method. We also find that 15% of countries use both methods or the Spanish method.⁵ The fact that one method is not overwhelmingly practiced worldwide is also evidence of the inconclusiveness of method ranking in treasury auctions.⁶

3 | THE ICELANDIC TREASURY MARKET

3.1 | Treasury auction procedures

In Iceland, the Ministry of Finance and the Central Bank of Iceland (CBI) have an agreement on treasury debt management. Act No. 43 has defined the relation between the Ministry of Finance and the CBI since 1990. The division of tasks between the two parties works as follows. On the one hand, the Ministry of Finance makes borrowing decisions based on its financial needs and determines the payment structure, maturity, and characteristics of the individual securities issued. The Ministry is also in charge of debt repayment and liquidity management. On the other hand, the CBI is in charge of the issuance protocol of the treasury securities and disseminates information to primary dealers and investors about the auction calendar. In addition, the CBI produces prospectus for newly issued securities and handles security listings with the Icelandic Securities Depository.

⁵The latter is a mixture of the uniform-price and the discriminatory method. See Álvarez and Mazón (2019) for a presentation of the Spanish method.

⁶In a cross-country analysis, Brenner, Galai, and Sade (2009) show that the uniform-price method is widespread across countries whose legal system follows common law, while the discriminatory method is more popular in countries following civil law. This result suggests that the auction method is, in many cases, more a legacy than the result of an analysis, in terms of efficiencies and revenues. Another interesting fact is that, over the last 30 years, only a few countries have switched auction methods, or at least have experimented over a few issuance cycles. To the best of our knowledge, only six countries have changed their auction methods. From the discriminatory method to the uniform-price method: Mexico (1990), the United States (1998), Korea (2000), Turkey (2001), and Poland (2012). From the uniform-price method to the discriminatory method: Mongolia (unknown date). This list of events comes from Monostori (2014).

In the Iceland Treasury market, securities are sold exclusively in auctions and there are no syndicated offerings. These auctions are organized on a regular basis. From 2000 to 2018, each month two auctions have been organized by the CBI on average. Three types of securities are issued: treasury bills, treasury bonds, and index-denominated treasury bonds. All securities share this common name structure: "RIKV yy mmdd" for T-bills, "RIKB yy mmdd" for T-bonds, and "RIKS yy mmdd" for index-link T-bonds. "yy", "mm" and "dd" stands for the year, month, and day of the maturity date, respectively. Treasury bonds are reissued multiple times in subsequent auctions, which are called reopening auctions.

The CBI communicates on expected issuance volume on an annual and quarterly basis, based on the government's estimated borrowing requirement and the auction calendar. Estimated volumes of issuance are disclosed under the section "Annual Prospects" of the Government Debt Management website www.lanamal.is. A few days before an auction, a document called "Terms of Invitation to Tender for Treasury Bills" for T-bills and "Terms of Invitation to Tender for Treasury Bonds" for T-bonds is disclosed to market participants. Hence, the supply of treasury securities is set in advance and not endogenous on bidding activities.

Bids can only be submitted to the CBI through the Bloomberg Auction System between 10:30 a.m. and 11 a.m. on the auction date. Both competitive and non-competitive bids are accepted. Bids are expressed on a price basis, with three decimals. Payments must be received by the CBI before 2 p.m. on the date of payment, and the securities are delivered in electronic form on the same day. This time schedule is the actual one. It has evolved a little between 2000 and 2018. For instance, in 2003, bids had to be sent between 10 a.m. and 11 a.m., while payments had to be made before 4 p.m. Once the auction winners are determined and awarded, the CBI notifies the stock exchange and the media about the auction results.

The PDs have exclusive access to the auctions. From 2000 to 2018, the list of PDs has always consisted of the largest Icelandic banks, and no foreign financial institution has ever been added to the list. The CBI writes a bilateral contract with each PD and renews it every year.⁷ One important aspect of the contract is mandatory participation: Each PD is obliged to submit bids at each auction for a minimum of 100 million Icelandic Krona (ISK) of face value. In compensation for this mandatory presence in the primary market, the CBI pays each dealer a commission, which takes the form of a small percentage of the winning bids and is deducted from the PD's payment for the purchase of the securities. There is no commission paid for auctions of T-bills. In addition, winning PDs have the right to purchase more quantity after the auction, up to the equivalent of 10% of the nominal value sold in the auction at the average winning price.

The decision to replace the discriminatory with the uniform-price method was taken on November 28, 2008, and announced on the same day to market participants through a press release. The first T-bond auction structured as a uniform-price auction took place on December 03, 2008. The change for T-bill auctions took place almost one year later, with the first auction under the uniform-price format organized on October 13, 2009. As to the motivations for the method change, the head of the Icelandic Government Debt Management told us that he was intrigued by the academic papers advocating the use of the uniform-price method. He also indicated that the switch was motivated by the attempt to increase the volume of transactions in the secondary market as well, and it would have been implemented even if the 2008 financial crisis had not happened.

3.2 | Auction data

Our sample of Icelandic Treasury auctions covers the period from January 2000 to March 2018. Each treasury security and primary dealer has a unique ID consistent throughout. We observe, for each auction, the date, the security issued, the settlement date, the primary dealer bids, the quantity asked per bid, and the quantity awarded per

⁷The list of general and bilateral agreements is available at <http://www.lanamal.is/EN/investors/primary-dealers-for-government-securities>.

TABLE 1 Auction-level summary statistics.

Securities	Full sample			Discriminatory method			Uniform-price method		
	T-bills	T-bonds	All	T-bills	T-bonds	All	T-bills	T-bonds	All
Number of auctions	246	270	516	95	91	186	151	179	330
Number of securities	182	19	201	85	7	92	97	15	112
Number of bidders	4	4	4	3	5	4	4	4	4
Number of bids	15	21	18	18	17	17	14	23	19

Note: This table shows summary statistics for our sample of Icelandic Treasury auctions between January 2000 and March 2018. The sample is split according to security types and auction methods. The first line indicates the total number of auctions organized by the Central Bank of Iceland, while the second line shows the total number of unique securities issued in the auctions. The difference between the total number of auctions and the total number of securities is explained by the existence of reopening cycles for the T-bonds. The third line shows the average number of bidders participating in each auction. The last line reports the average total number of bids received in an auction.

bid. Summary statistics are reported in Table 1. The total number of auctions is 516. There are 246 (270) auctions of T-bills (T-bonds).⁸ When the sample is split according to the auction method, there are 95 auctions of T-bills organized under the discriminatory method and 151 auctions of T-bills organized under the uniform-price method. For the T-bond market, there are 91 auctions organized under the discriminatory method and 179 auctions organized under the uniform-price method. The first important characteristic in our sample is the low number of T-bonds, which is explained by the CBI's intensive use of reopening cycles. The 270 auctions of T-bonds represent the issuance of only 19 unique securities. RIKB 07 0209, RIKB 10 0317, RIKB 13 0517, RIKB 20 0205, and RIKB 31 0124 were re-issued more than 20 times. The second important characteristic of our sample is the low number of PDs. On average, there are four PDs participating in the Icelandic Treasury auctions.⁹

3.3 | Secondary market data

The secondary market data is from Nasdaq Nordic, which is the subsidiary of Nasdaq, Inc., operating at marketplaces for financial securities in the Nordic and Baltic regions of Europe. The data includes daily closing prices and trading volumes for most of the bills and bonds issued by the Icelandic government. By merging our auction sample with Nasdaq Nordic data, we obtain a final sample of 385 auctions (75% of the 516 auctions) for which underpricing is computable. Among these auctions, there are 262 (69.1%) security-auction day observations that find exact matches with secondary market data. For these observations, the security is traded in the secondary market on the same day as the auction. For the other 123 security-auction day observations, the first recorded transaction occurs in the secondary market one day, two days, and up to five days after the auction takes place. We exclude 131 (516-385) security-auction day observations from our final sample as the first transaction in the secondary market occurs more than five days after the auction. The numbers of observations split across auction methods are as follows: 153 and 232 for the discriminatory and uniform-price methods, respectively.

⁸The index-linked treasury securities are not issued on a regular basis and turn out to be very illiquid in the secondary market, we remove these securities from our analysis.

⁹To compare to another Nordic country, there are between 5 and 10 PDs in the sample of Keloharju, Nyborg, and Rydqvist (2005) concerning Finland treasury auctions between 1992 and 2000.

3.4 | Government Debt Management Data

A detailed description of commission data, lending facility data, and trading lots is provided in Appendix 1.

4 | BIDDING BEHAVIOR AND AUCTION UNDERPRICING

4.1 | Annual statistics

In Figure 2, we exhibit several statistics on auctions, security supply, PDs, and bids, by year and by security type. First, the number of auctions changes over time. There are progressively more and more auctions of T-bills and T-bonds: between five and 10 for the discriminatory method and between 10 and 25 under the uniform-price method. Auction size also increases. For the T-bill market, peaks are observed in 2008, 2009, and 2010, consistent with the borrowing needs of the Icelandic government. In Iceland, the outstanding amounts are historically larger for short-term treasury securities than for long-term ones, and the T-bond auction size catches up to the T-bill auction size only in 2013. We can also observe that the number of PDs participating in the auctions is low and stable over time: At least (at most) three (five) PDs participate in a treasury auction. One explanation is that only financial institutions with Icelandic nationality are authorized to participate.

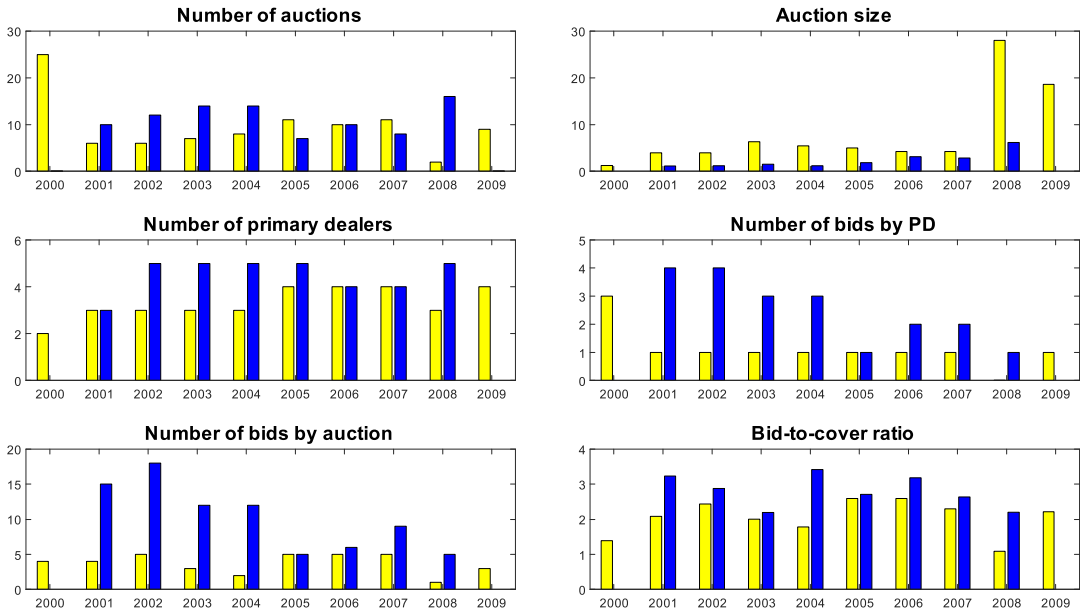
We find an interesting pattern in bidding behavior over time: The number of bids is larger for the T-bill market between 2000 and 2010, but it is larger for the T-bond market between 2011 and 2017, suggesting an asymmetric effect of the method change on T-bills and T-bonds. There is relatively more competition between PDs in T-bond auctions under the new format. The bid-to-cover ratio, a widespread measure of auction performance, defined as the total amount asked by PDs over the total supply, is lower on average for the 2011–2017 period, for both types of treasury securities. This is due to the fact that the supply of treasury securities is larger on average for the 2011–2017 period.

4.2 | Bidder-level statistics

We do not know the true identities of the PDs but our dataset contains consistent bidder IDs over time. These are: PD_1, PD_2, PD_3, PD_4, PD_5, PD_6, PD_7 and PD_8. The arrival of PD8 among the primary dealers and the withdrawal of PD_1, PD_5, and PD_6 coincides with the setup of the new method. Summary statistics on PD bidding behavior and performance under the discriminatory method are reported in Table 2. We can observe some heterogeneity in terms of bidding intensity and the amount awarded. For instance, for T-bond auctions during the discriminatory method period, PD_6 submits only one bid on average and is awarded securities in only eight auctions, that is 24% of the auctions in which the PD participates. The quantity awarded, conditional on winning, to this PD is 9.1% on average. On the opposite side of the panel, PD_3 turns out to be very active: This bidder submits five bids on average, wins in 74 auctions (84%), and, conditional on winning, is awarded 46.8%. Interestingly, strong bidding intensity does not necessarily lead to large award amounts. For instance, PD_7 submits on average 9 bids on the T-bill market, which is very high compared to the other PDs. However, that PD is awarded only 19.7% of the securities, on average, and conditioned on winning the auction. Unlike PD_7, PD_1 submits only four bids on average but is always awarded the auctions he participated in. On average, he is awarded 24.8% of the supply of securities.

PD bidding behavior and performance under the uniform-price method are reported in Table 3. We observe that there are more auctions and fewer auction participants since PD_1, PD_5, and PD_6 have ceased participating. Each PD seems to be very active except for PD_8, who participates in less than half of the auctions organized, submitting one (three) bid only on average during T-bill (T-bond) auctions. However,

Panel A: Discriminatory method



Panel B: Uniform-price method

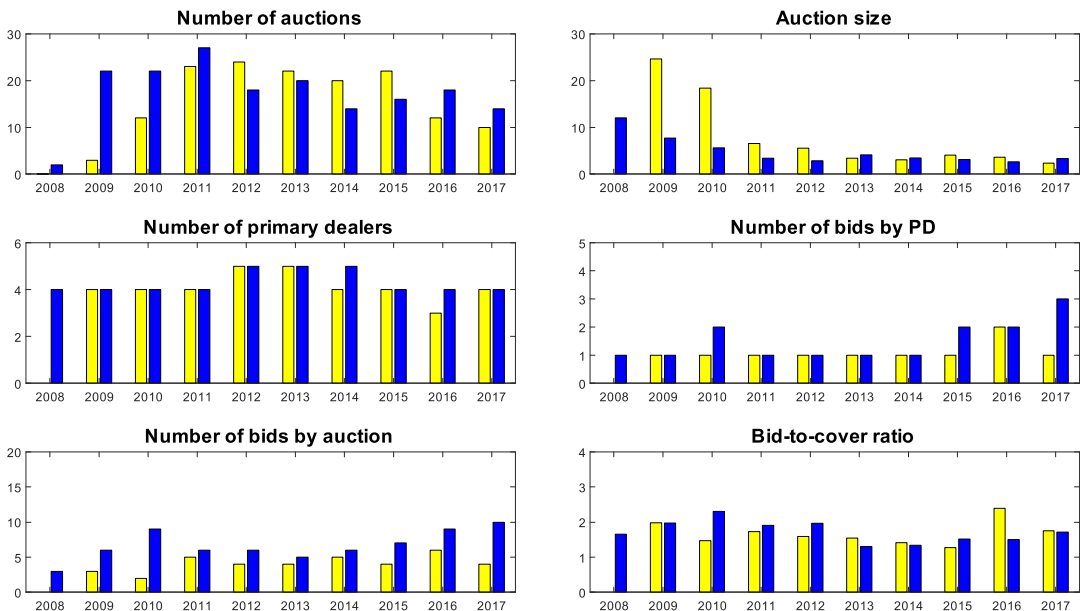


FIGURE 2 Auction statistics over 2000–2018. *Panel A: Discriminatory method. Panel B: Uniform-price method.* Top left: Number of treasury auctions organized each year. Top right: Average auction size, in billions of nominal value of treasury securities. Middle left: Number of primary dealers participating in auctions. Middle right: Number of bids by PD for one billion of nominal value. Bottom left: Total number of bids scaled by one billion nominal value. Bottom right: Bid-to-cover ratio. Numbers are averages across auctions. Yellow: Icelandic Treasury bills, Blue: Icelandic Treasury bonds. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 2 Primary dealer activity under the discriminatory method period.

Bidder ID	PD_1	PD_2	PD_3	PD_4	PD_5	PD_6	PD_7
<i>Treasury bills</i>							
1) Number of auctions	9	88	37	89	42		38
2) Number of bids	4	6	6	5	4		9
3) Number of winning auctions	9	64	32	74	30		26
4) Winning auctions (%)	100	72.7	86.5	83.1	71.4		68.4
5) Demand (ISK)	1.2	4.6	5.4	3.3	1.6		2.7
6) Demand-to-supply (%)	29.5	71.2	86.7	72.3	39.1		35.7
7) Awards (ISK)	1.0	3.0	2.6	2.2	1.0		1.7
8) Awards-to-supply (%)	24.8	40.1	38.5	48.1	26.9		19.7
<i>Treasury bonds</i>							
1) Number of auctions	22	76	88	89	59	33	43
2) Number of bids	4	4	5	3	2	1	5
3) Number of winning auctions	11	43	74	67	35	8	22
4) Winning auctions (%)	50	56.6	84.1	75.3	59.3	24.2	51.2
5) Demand (ISK)	0.5	0.9	2.4	1.8	0.4	0.1	1.1
6) Demand-to-supply (%)	49.8	42.6	124.7	65.6	25.5	8.5	38.6
7) Awards (ISK)	0.2	0.6	1.1	1.2	0.3	0.1	0.5
8) Awards-to-supply (%)	18.7	21.7	46.8	40.4	16.5	9.1	15.5

Note: 1) The number of auctions in which the primary dealer has sent one bid at least. 2) The rounded average number of bids sent by the primary dealer. 3) The number of auctions where the primary dealer wins one treasury security at least. 4) The number of winning auctions divided by the total number of auctions. 5) The average demand size for treasury securities in billions of Icelandic Krona (ISK). 6) The average demand-to-supply of treasury securities. 7) The average award size for treasury securities in billions of Icelandic Krona (ISK). 8) The average award-to-supply of treasury securities.

that PD is awarded 59.7% (20.2%) on average per win. PD_2 is the largest winner, with an average of 60.5% (35.9%) of the total T-bill (T-bond) amount awarded.

4.3 | Underpricing

To be consistent with the literature, we use the following notation: i for auction, j for bidder, and k for bid. Each bidder j sends one or several price-quantity pairs $\{b_{i,j,k}; q_{i,j,k}\}$ to the auctioneer, who ranks the bids in price descending order. Then, the treasury securities are awarded to the bids following the descending order, up to cumulated awards equal to the intended volume. The winning pairs are denoted $\{b_{i,j,k}^*, q_{i,j,k}^*\}$. Under the discriminatory method, each bidder pays $b_{i,j,k}^*$ to receive $q_{i,j,k}^*$. Under the uniform-price method, each bidder pays $\inf\{b_{i,j,k}^*\}$ instead of $b_{i,j,k}^*$ to receive $q_{i,j,k}^*$. That is, all bidders pay a unique price, which is the lowest bid at which demand equals supply. It is commonly called the market clearing price or stop-out price.

Under the discriminatory method, underpricing is measured as follows:

$$U_{DISCR}^i = \frac{P_i - \sum w_{i,j,k}^* \times b_{i,j,k}^*}{\sum w_{i,j,k}^* \times b_{i,j,k}^*} \tag{1}$$

TABLE 3 Primary dealer activity under the uniform-price method period.

Primary dealer ID	PD_2	PD_3	PD_4	PD_7	PD_8
<i>Treasury bills</i>					
1) Number of auctions	141	144	142	132	61
2) Number of bids	4	4	4	2	1
3) Number of winning auctions	127	134	117	39	13
4) Winning auctions (%)	90.1	93.1	82.4	29.5	21.3
5) Demand (ISK)	4.5	2.1	1.3	0.7	0.8
6) Demand-to-supply (%)	89.3	32	28.3	13.1	18.8
7) Awards (ISK)	3.4	1.9	1.1	1	2.9
8) Awards-to-supply (%)	60.5	28.9	19.8	11.4	59.7
<i>Treasury bonds</i>					
1) Number of auctions	176	168	174	176	61
2) Number of bids	5	4	6	8	3
3) Number of winning auctions	161	128	147	159	37
4) Winning auctions (%)	91.5	76.2	84.5	90.3	60.7
5) Demand (ISK)	2.4	1.0	1.3	1.6	0.8
6) Demand-to-supply (%)	63.9	28.6	34.4	43.5	24.1
7) Awards (ISK)	1.6	0.8	1.1	1.2	0.9
8) Awards-to-supply (%)	35.9	21	25.6	30.9	20.2

Note: 1) The number of auctions in which the primary dealer has sent one bid at least. 2) The rounded average number of bids sent by the primary dealer. 3) The number of auctions where the primary dealer wins one treasury security at least. 4) The number of winning auctions divided by the total number of auctions. 5) The average demand size for treasury securities in billions of Icelandic Krona (ISK). 6) The average demand-to-supply of treasury securities. 7) The average award size for treasury securities in billions of Icelandic Krona (ISK). 8) The average award-to-supply of treasury securities.

where

$$w_{i,j,k}^* = \frac{q_{i,j,k}^*}{\sum q_{i,j,k}^*} \quad (2)$$

P_i is the price observed in the secondary market after auction i . Under the discriminatory method, underpricing is measured as the scaled difference between the secondary market price observed post-auction and the quantity-weighted average winning price. Under the uniform-price method, underpricing is measured as follows:

$$U_{UNIF}^i = \frac{P_i - \inf\{b_{i,j,k}^*\}}{\inf\{b_{i,j,k}^*\}} \quad (3)$$

Table 4 reports underpricing levels across auction methods. The average underpricing in price space (yield space) is 7.86 (15.14) basis points under the discriminatory method and -0.67 (-12.98) basis points under the uniform-price method. The median numbers are 4.73 (2.99) basis points and 2.98 (1.14) basis points, respectively. A non-parametric test on the medians produces a z-value of 2.03 (3.64), showing that the decline is significant at the

TABLE 4 Underpricing statistics.

	Full period			Discriminatory method period			Uniform-price method period			Rank
	N	Mean	Median	N	Mean	Median	N	Mean	Median	Sum test z-value
Price space	385	2.72	3.78	153	7.86	4.73	232	-0.67	2.98	2.03**
Yield space	385	-1.81	1.72	153	15.14	2.99	232	-12.98	1.14	3.64***

Note: This table reports several descriptive statistics on the level of underpricing under the two auction methods. We define underpricing as the daily closing price of the treasury security recorded in the secondary market price minus the auction outcome price. The latter is the quantity-weighted average winning price under the discriminatory method and the stop-out price under the uniform-price method. If there is no closing price in the secondary market on auction day, we use the closest available closing price within the next five trading days instead. Yields space means that yields replace prices for the calculation of underpricing. In the last column, the z-values are derived from Wilcoxon Rank Sum tests. *, **, *** denote statistical significance at the 10%, 5%, and 1%, respectively.

5% level. Underpricing for Icelandic Treasury securities seems to be fairly close to what is documented for other countries. Goldreich (2007), working on a set of 283 auctions of US Treasury securities over the 1991–2000 period, finds that underpricing is 3.5 basis points for discriminatory auctions on average while underpricing in uniform-price auctions is 1.3 basis points. Bjonnes (2001), working on a set of Norway Treasury securities, finds 3.6 basis points for T-bill issuance under the discriminatory method and 13.3 basis points for T-bonds issued under the uniform-price method. Keloharju, et al. (2005) find an average of 4.1 basis points on the Finnish Treasury security market under the uniform-price method.¹⁰

5 | EVIDENCE FROM MULTIVARIATE REGRESSIONS

5.1 | The model

To assess whether the change in auction method is associated with a significant change in underpricing level, we estimate the following model using ordinary least square regression:

$$UNDERPRICING_i = \beta_0 + \beta_1 \times AUCTION_METHOD_i + \delta' \times EXPLANATORY_i + \varepsilon_i \tag{4}$$

where

$$UNDERPRICING_i = \begin{cases} U_{DISCR}^i & \text{if } AUCTION_METHOD_i = 0 \\ U_{UNIF}^i & \text{if } AUCTION_METHOD_i = 1 \end{cases} \tag{5}$$

The subscript i denotes the auction. U_{DISCR}^i and U_{UNIF}^i are defined in (1) and (3) and expressed in basis points. $AUCTION_METHOD$ is a dummy variable equal to zero (one) for auctions occurring under the discriminatory (uniform-price) method. $EXPLANATORY$ is a large set of explanatory variables, including auction and security characteristics, obligations and privileges imposed on PDs, and financial market conditions.

$AUCTION_SIZE$ is the natural logarithm of the number of treasury securities issued in the auction. The security characteristics are a reopening cycle dummy equal to one if the security has been issued in the past, zero otherwise

¹⁰We note that a direct comparison of underpricing across country studies is far from rigor, especially due to the fact that most studies have no secondary market transaction prices and therefore rely on bid, ask, or mid prices.

(*REOPCYCLES*), and another dummy variable equal to one for T-bonds and zero otherwise (*BILL_BOND_DUM*). In addition, we control for some factors related to the obligations, rights, and privileges the CBI imposes on the PDs. These variables are the number of accredited PDs (*NB_BIDDERS*), the minimum trading lot size in the secondary market (*MIN_TRADING_LOT*), the maximum loan amount at the disposal of the PDs (*LEND_FAC*), the commissions paid to primary dealers (*COMMISSION*), and the option to buy additional securities in the auction (*OPTION_DUM*). As for auction performance variables, we use bid-to-cover ratio (*BID_TO_COVER*), measured as the total demand for treasury securities in % of the supply, the dispersion (*BID_DISPERSION*), which captures the heterogeneity of security valuation among PDs, and award concentrations (*AWARD_CONCEN*), measured as the number of securities concentrated in the hands of the largest winner, in % of the auction supply.¹¹ Our regressions also incorporate financial market conditions and a measure of the Icelandic sovereign credit risk. We compute the volatility of treasury bonds (*BOND_VOL*) with our secondary market data, and we measure the performance (*ICEX_RTN*) and volatility (*ICEX_VOL*) of the equity market with the Icelandic Stock Market (ICEX) index returns. As a proxy for the sovereign credit risk, we use the difference between the 5-year Icelandic bond yield benchmark and the 5-year German bond yield benchmark (*5Y_SPREAD*).¹² Appendix 2 provides a summary of all variable definitions.

5.2 | Empirical results

Table 5 contains the main regression results.¹³ If the uniform-price method leads to a lower underpricing level, the coefficient associated with the auction method dummy variable should be negative. The second column shows the result of the univariate regression. β_1 is negative (−0.09) and statistically significant at the 1% level. Columns 3 to 9 show multivariate regression results. β_1 turns out to be negative and statistically significant at 5% in six models out of seven and significant at 10% in all specifications. When we include all explanatory variables together, β_1 remains negative (−0.17) and significant at the 5% level. Regression results confirm the preliminary evidence provided by the non-parametric tests. The auction method change from the discriminatory to the uniform-price one is associated with a statistically significant reduction in underpricing.

Beyond highlighting the importance of the auction method, the regressions also bring to light the connections between underpricing and some explanatory variables, which deserve to be analyzed and commented on. First, the positive and significant relationship between *AUCTION_SIZE* and underpricing is consistent with the risk aversion theory. Under this theory, larger auctions involve more aggregate security risk to be shared among the primary dealers (Nyborg, Rydqvist, and Sundaresan, 2002), who respond by shading their bids. More significant discounts produce a lower auction outcome price, which decreases underpricing. We also observe that the impact of *MIN_TRADING_LOT* on underpricing is significant and positive. Increasing the minimum trading lot generates additional inventory management costs and risk, for which primary dealers require compensation. This compensation takes the form of an increasing underpricing level. The positive and statistically significant association of *AUCTION_SIZE* and *MIN_TRADING_LOT* with *UNDERPRICING* may be a consequence of the limited risk-bearing capacity of the financial institutions that participate in the Icelandic treasury market, as PD in the primary market and marker-maker in the secondary market. The limited risk-bearing capacity of the PDs, recently

¹¹A PD awarded with a substantial fraction of the securities issued has a dominant position in the secondary market. This outcome might be unintentional or intentional. Under this latter scenario, the PD might seek to establish a short-squeeze position, raising secondary market prices. For an analysis of the influence of short squeezes on bidding behavior and underpricing, see Rydqvist and Wu (2016).

¹²The credit spread might influence the auction outcome in two ways. First, a change in the sovereign credit risk impacts the valuation of treasury securities. Second, our measure may capture, to some extent, a change in the credit risk of the banking sector because the sovereign credit risk correlates with the credit risk of the national banks. The PDs may behave differently in the auctions if their credit risk evolves.

¹³The number of observations changes a little across regressions. Column 5: We cannot calculate bid dispersion for four auctions because there is one single bidder. Column 7: We lose eleven auctions over March 2000 - June 2001 as commission data are unknown before June 2001. Column 8: We lose 16 auctions because the credit spread variable is unknown before October 2001. Column 9: We lose twenty auctions when introducing all explanatory variables simultaneously.

TABLE 5 Regression results.

AUCTION_METHOD	-0.085*** (0.029)	-0.154*** (0.059)	-0.131*** (0.035)	-0.075** (0.031)	-0.112*** (0.035)	-0.064* -0.033	-0.082** (0.039)	-0.170** (0.071)
AUCTION_SIZE		0.107*** (0.021)						0.132*** (0.026)
NB_BIDDERS		-0.029 (0.023)						-0.013 (0.031)
REOP_CYCL_DUM		0.246*** (0.064)						0.100 (0.083)
OPTION_DUM		-0.036 (0.057)						0.031 (0.072)
BILL_BOND_DUM			0.195*** (0.044)					0.304*** (0.075)
BID_DISPERSION				2.614 (6.674)				-3.982 (6.661)
BID_TO_COVER				0.028* (0.017)				0.055*** (0.014)
AWARD_CONCEN				-0.291*** (0.102)				-0.199** (0.082)
MIN_TRADING_LOT					0.001 (0.001)			0.002*** (0.001)
LEND_FAC					0.049*** (0.011)			0.003 (0.015)
COMMISSION						-0.079** -0.037		-0.210** (0.084)
TREA_BOND_VOL							0.005 (0.005)	0.007 (0.004)
5Y_SPREAD							-0.006 (0.015)	-0.002 (0.016)
CONTROL_VAR	No	No	No	No	No	No	Yes	Yes
Number of observations	385	385	385	381	380	374	369	365
Adjusted R-squared	1.4	11.5	8.7	3.9	6.9	1.9	2.2	25.4

Note: We regress underpricing on the auction method and a set of explanatory variables. We define underpricing as the daily closing price of the treasury security recorded in the secondary market minus the auction outcome price, divided by the auction outcome price. The latter is the quantity-weighted average winning price under the discriminatory method and the stop-out price under the uniform-price method. If there is no closing price in the secondary market on the auction day, we use the closest closing price within the next five trading days instead. Appendix 2 provides the explanatory variable definitions. Standard errors, reported below the coefficients, are adjusted for data heteroscedasticity with the Huber-White correction. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

documented in Lou, Yan, and Zhang (2013) and Amin and Tédongap (2023), is responsible for a particular pattern of the secondary market price around the auction. We analyze this pattern under the two auction methods in the next section.

Second, *AWARD_CONCEN* is a highly significant variable with a negative coefficient sign. We explain its connection to underpricing by the following mechanism: a PD will bid aggressively in the auction if he is willing to acquire a large portion of the securities issued, which increases the auction outcome price and reduces underpricing. Suppose a concentration of the issue is in the hands of a single PD, leading him to abuse its market power. In that case, we should find a positive relationship between *AWARD_CONCEN* and underpricing. However, our result rejects this possibility. In the Icelandic treasury market, PDs do not seek to establish a dominant market position or trigger short-squeeze. The market-making rules established by the CBI, signed by the PDs, and updated regularly, seem to work well in preventing these behaviors.

Third, the treasury bond market volatility (*TREA_BOND_VOL*), the Icelandic-German bond spread (*5Y_SPREAD*), and the measures of financial market conditions (*CONTROL_VAR*) are non-significant. This result suggests that underpricing in the Icelandic treasury market is primarily affected by factors related to the auction itself rather than global or local market conditions. This result is surprising because bond volatility is a significant variable in explaining underpricing in Goldreich (2007). A critical organizational difference between the two treasury markets may be responsible for this divergence: The CBI does not allow foreign financial institutions to participate in the auctions, while the US Treasury does. These foreign financial institutions might be more sensitive to market conditions than domestic banks, and this sensitivity influences the way they behave in the auction. Consequently, underpricing is more likely related to market factors when these foreign financial institutions participate in the auction.

The fourth interesting result is the significant negative impact of the commission paid to primary dealers (*COMMISSION*) on underpricing. More commissions are associated with a lower underpricing level. A possible explanation is that commissions are a substitute for the gain made by the PDs with underpricing. We discuss the measurement of issuance costs when PDs are paid in the next section. However, how a remuneration policy set by a sovereign issuer influences the way PDs behave or affect their market-making strategy is beyond the scope of this study, and we will address these research questions in another paper. Overall, the statistical significance of the variables *MIN_TRADING_LOT* and *COMMISSION* shows that the treasury rules that framed the primary dealer activities are essential for explaining underpricing. Future researchers on bidding behavior in treasury auctions should incorporate the specific rules set by the treasury agency.

5.3 | Robustness checks

Underpricing can be measured in price or yield space; both measures are standard in the literature. To obtain consistent signs, underpricing in yield space is the difference between the auction outcome yield and the secondary market yield. It is the incremental return a market participant earns by winning a security in the auction relative to purchasing the same security in the secondary market. The auction outcome yield is the quantity-weighted average winning yield under the discriminatory method and the stop-out yield under the uniform-price method. We calculate secondary market yields with the Newton-Raphson method applied to daily closing prices. Since the relationship between prices and yields is non-linear, the model with underpricing in yield space may produce coefficients and standard errors different from what we obtain in the previous regressions.

We also implement a correction to underpricing. Given the daily frequency of our market data, the secondary market price used to calculate underpricing is the closest end-of-day price if no such price exists on the auction day. If any information relevant to value treasury securities arrives between the auction day and the next trading day,

the security price will incorporate this information. Underpricing will be biased as it contains a component related to information. To neutralize this component, we propose the following adjustment:

$$U_{DISCR}^* = U_{DISCR} - \frac{P_{a+t}^{MS} - P_a^{MS}}{P_a^{MS}} \quad (6)$$

for the discriminatory method and

$$U_{UNIF}^* = U_{UNIF} - \frac{P_{a+t}^{MS} - P_a^{MS}}{P_a^{MS}} \quad (7)$$

for the uniform-price method, where P_a^{MS} is the daily closing price of a matching security on the auction date, while P_{a+1}^{MS} is the closing price recorded t days after. Hamao and Jegadeesh (2002) propose a similar adjustment. t depends on the availability of the secondary market price for the security. For instance, if the secondary market price of this security is recorded four days after the auction day, $t = 4$, so the percentage price change of the matching security spans four days after the auction. A treasury bill (bond) is matched with another treasury bill (bond) with a comparable maturity.

We run eight regressions by combining the following criteria: univariate or multivariate, presence or absence of matching securities, and price space or yield space. Table 6 shows the regression results. The variable *AUCTION_METHOD* is negative and significant at the 10% level for all specifications and at 5% in six regressions. It confirms the evidence that underpricing is lower under the uniform-price method, and this result holds regardless of how the dependent variable is measured.

Iceland experienced financial turmoil starting in September 2008 when the interbank lending market was reduced drastically following the collapse of Lehman Brothers in the United States.¹⁴ Given that the CBI implemented the uniform-price auction method in October 2008, the financial crisis could be a confounding factor generating a spurious association between the auction method and underpricing. In order to convince that the financial crisis does not drive our results, we rerun the regressions in a sample that excludes the 104 auctions that took place from September 2008 (the banking crisis started) to August 2011 (the international bailout support program for Iceland officially ended). Table 7 reports these regression results. We observe that the auction method dummy remains significant at the 5% level in all regressions. We conclude that the Icelandic financial crisis does not induce the significant relationship between the auction method and underpricing.

The specific bidding strategy of each PD could be persistent through time, and bid correlations over successive auctions could generate autocorrelated residuals in our model. We compute the Durbin-Watson statistics for each regression in Table 5 to verify the autocorrelation's severity. We reject the null hypothesis of the absence of autocorrelation for three regressions out of eight. In unreported results, we calculate the Newey-West standard errors and find that the auction method dummy remains significant at the 10% level in the three cases.

6 | DISCUSSIONS

6.1 | Auction method, underpricing and the winner's curse

Our dataset contains all winning and losing bid-quantity pairs, which allows us to test auction theories and link bidding behavior to the auction method. The winner's curse is a popular theory that explains bidder behaviors in the auction literature. The auction winners are the most optimistic bidders concerning the asset, and they are cursed

¹⁴For an analysis of the causes of the Icelandic financial crisis, see Aliber and Zoega (2011).

TABLE 6 Regression results- robustness measures.

Dependent variable	Price space		Yield space	
	Raw underpricing	Adjusted underpricing	Raw underpricing	Adjusted underpricing
AUCTION_METHOD	-0.085*** (0.029)	-0.170** (0.071)	-0.058* (0.033)	-0.147* (0.081)
			-0.281*** (0.056)	-0.528*** (0.176)
				-0.124** (0.062)
				-0.329** (0.144)
AUCTION_SIZE	0.132*** (0.026)	0.145*** (0.028)	0.142*** (0.053)	0.182*** (0.057)
NB_BIDDERS	-0.013 (0.031)	-0.036 (0.034)	-0.048 (0.065)	-0.085 (0.071)
REOP_CYCL_DUM	0.100 (0.083)	0.114 (0.082)	0.213 (0.133)	0.354** (0.140)
OPTION_DUM	0.031 (0.072)	0.049 (0.075)	0.079 (0.176)	0.054 (0.163)
BILL_BOND_DUM	0.304*** (0.075)	0.389*** (0.080)	0.352*** (0.135)	0.515*** (0.141)
BID_DISPERSION	-3.982 (6.661)	4.366 (10.099)	-12.680 (7.978)	-3.978 (6.200)
BID_TO_COVER	0.055*** (0.014)	0.065*** (0.016)	0.026 (0.024)	0.050* (0.027)
AWARD_CONCEN	-0.199** (0.082)	-0.314*** (0.093)	-0.364** (0.165)	-0.505*** (0.192)
MIN_TRADING_LOT	0.002*** (0.001)	0.002** (0.001)	0.005*** (0.002)	0.003 (0.002)

(Continues)

TABLE 6 (Continued)

Dependent variable	Price space		Yield space	
	Raw underpricing	Adjusted underpricing	Raw underpricing	Adjusted underpricing
LEND_FAC	0.003 (0.015)	-0.003 (0.016)	0.019 (0.039)	0.018 (0.037)
COMMISSION	-0.210** (0.084)	-0.189** (0.089)	-0.428** (0.186)	-0.255 (0.235)
TREA_BOND_VOL	0.007 (0.004)	0.007* (0.004)	0.005 (0.005)	0.010* (0.005)
5Y_SPREAD	-0.002 (0.016)	-0.011 (0.019)	0.038 (0.030)	-0.002 (0.035)
CONTROL_VAR	No	Yes	No	Yes
Number of observations	385	365	385	365
Adjusted R-squared	1.4	25.4	5.1	23.8

Note: We regress underpricing on the auction method and a set of explanatory variables. In columns 2 to 5 (price space), underpricing is the daily closing price of the treasury security recorded in the secondary market minus the auction outcome price, divided by the auction outcome price. The latter is the quantity-weighted average winning price under the discriminatory method and the stop-out price under the uniform-price method. If there is no closing price in the secondary market on the auction day, we use the closest closing price within the next five trading days instead. We adjust underpricing in columns 4 and 5 by subtracting the matching security daily price variation. In columns 6 to 9, yields replace prices for the calculation of underpricing. Yields are derived with the Newton-Raphson method. Appendix 2 provides the explanatory variable definitions. Standard errors, reported below the coefficients, are adjusted for data heteroscedasticity with the Huber-White correction. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

TABLE 7 Regression results- excluding financial crisis period.

Dependent variable	Price space		Yield space					
	Raw underpricing	Adjusted underpricing	Raw underpricing	Adjusted underpricing				
AUCTION_METHOD	-0.143*** (0.039)	-0.245** (0.121)	-0.131*** (0.043)	-0.306** (0.122)	-0.325*** (0.072)	-0.650** (0.276)	-0.191** (0.085)	-0.797*** (0.274)
AUCTION_SIZE	0.127*** (0.035)	0.151*** (0.037)	0.118* (0.070)	0.172** (0.070)				
NB_BIDDERS	-0.015 (0.033)	-0.025 (0.034)	-0.027 (0.066)	-0.081 (0.070)				
REOP_CYCL_DUM	0.126 (0.111)	0.211* (0.110)	0.320* (0.180)	0.538*** (0.200)				
OPTION_DUM	-0.070 (0.105)	-0.026 (0.105)	-0.146 (0.274)	-0.024 (0.250)				
BILL_BOND_DUM	0.364*** (0.092)	0.430*** (0.097)	0.519*** (0.179)	0.739*** (0.195)				
BID_DISPERSION	-8.207 (8.624)	-16.493* (8.663)	-14.768 (9.355)	-12.974 (9.956)				
BID_TO_COVER	0.043* (0.023)	0.060** (0.026)	0.012 (0.044)	0.038 (0.044)				
AWARD_CONCEN	-0.170* (0.095)	-0.240** (0.107)	-0.248 (0.192)	-0.360* (0.218)				
MIN_TRADING_LOT	0.002 (0.001)	0.002 (0.001)	0.007*** (0.003)	0.007** (0.003)				

(Continues)

TABLE 7 (Continued)

Dependent variable	Price space		Yield space	
	Raw underpricing	Adjusted underpricing	Raw underpricing	Adjusted underpricing
LEND_FAC	-0.003 (0.021)	-0.006 (0.022)	0.023 (0.052)	-0.004 (0.052)
COMMISSION	-0.257** (0.129)	-0.284** (0.138)	-0.658** (0.296)	-0.596* (0.352)
TREA_BOND_VOL	-0.008 (0.009)	-0.009 (0.009)	-0.009 (0.018)	-0.011 (0.018)
5Y_SPREAD	0.066 (0.046)	0.082* (0.048)	0.166** (0.085)	0.196** (0.092)
CONTROL_VAR	No	Yes	No	Yes
Number of observations	281	262	281	263
Adjusted R-squared	4.3	23.4	6.6	29.3

Note: We regress underpricing on the auction method and a set of explanatory variables. The sample of auctions excludes the period September 2008 - August 2011. In columns 2 to 5 (price space), underpricing is the daily closing price of the treasury security recorded in the secondary market minus the auction outcome price, divided by the auction outcome price. The latter is the quantity-weighted average winning price under the discriminatory method and the stop-out price under the uniform-price method. If there is no closing price in the secondary market on the auction day, we use the closest closing price within the next five trading days instead. We adjust underpricing in columns 4 and 5 by subtracting the matching security daily price variation. In columns 6 to 9, underpricing is calculated with yields instead of prices. Appendix 2 provides the explanatory variable definitions. Standard errors, reported below the coefficients, are adjusted for data heteroscedasticity with the Huber-White correction. *, **, *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

because they have overestimated its value and overpaid for it. Bidders tend to bid below their estimation, a phenomenon called bid shading, to avoid ending up in this situation.

The winner's curse is critical in treasury markets because its severity directly affects Treasury revenues. The auction design might influence how the bidders adjust their strategy to avoid the winner's curse. In Milgrom and Weber (1982), the second-price auction is superior to the first-price auction in terms of revenue generated because the winner's curse is alleviated under the second-price auction. Milgrom (1989) and Bikhchandani and Huang (1993) study the winner's curse theory in the context of treasury markets and document that the response of the bidders to the winner's curse is different under the two auction methods. The uniform-price method should generate more aggressive bids because the winners pay a single price equal to the lowest winning bid, which limits the risk of overpaying compared to a discriminatory design. In Nyborg and Sundaresan (1996), auctions organized under the uniform method exhibit more pre-auction information release than those organized under the discriminatory method, lowering the winner's curse. Nyborg, et al. (2002) provide evidence that the winner's curse is an essential concern for the bidders of the Swedish treasury market, where the security issuance process follows the discriminatory method. Rational bidders account for the winner's curse by submitting bids at prices lower than their estimates, and they further reduce their bids when signals are imprecise.

We investigate if the winner's curse channel might be responsible for the decline in underpricing following the adoption of the uniform-price method by the CBI. We capture the bidders' behaviors with three bidder-level measures: i. bid dispersion, ii. the fraction of aggressive bids, and iii. bid premium. Bid dispersion is the quantity-weighted standard deviation of bids. The second measure is the fraction of bids posted above the closest closing day price before the auction starts. The third measure is the difference between the quantity-weighted average price and the closest closing day price before the auction starts. If the uniform-price method alleviates the winner's curse, we should observe the placement of more aggressive bids under this method. Our two measures capturing bidding aggressiveness should be positively associated with our auction method dummy after controlling for auction characteristics, security characteristics, and financial market conditions. We also expect that bid dispersion will correlate negatively with the auction method dummy.

Table 8 shows the multivariate regression results. Consistent with the predictions, we find that bid dispersion is negatively associated with the auction method dummy (columns 2 and 3) while bidding aggressiveness is positively associated with the dummy (columns 4 to 7). The coefficients are all significant at the 1% level.¹⁵ We conclude that the PDs increase the concentration of their bids and, simultaneously, tend to place more aggressive bids under the uniform-price method. This difference in bidding behavior is consistent with the theoretical predictions of Milgrom (1989) and Bikhchandani and Huang (1993). The uniform-price method, using a single outcome price equal to the lowest winning bid, fosters bidding aggressiveness from the PDs. At the end of the auction, the outcome price is higher relative to the secondary market price, lowering underpricing.

6.2 | Auction method and price impact

Lou, et al. (2013) analyze secondary market prices around the auction day in the context of US Treasury securities. Lou, et al. (2013) argue that if a treasury department could issue toward secondary market participants directly, he would issue a fraction of the whole amount of securities in a window of trading days instead of reselling the whole issuance on a single day in order to limit the price impact. Therefore, researchers should consider the price evolution around the auction when calculating the shortfall that underpricing represents for a government. Lou, et al. (2013) estimate the amount of money the US government could have saved were it able to issue the treasury securities a few days around the auction day.

¹⁵In unreported results, the coefficients remain significant when we replace White robust standard errors with Newey-West ones.

TABLE 8 Bidder-level regressions.

Dependent variable	Bid dispersion		Fraction of aggressive bids		Bid premium	
AUCTION_METHOD	-0.041*** (0.015)	-0.128*** (0.038)	0.231*** (0.061)	0.522*** (0.197)	0.794*** (0.128)	1.608*** (0.406)
AUCTION_SIZE		0.024*** (0.008)		0.059** (0.030)		0.188 (0.130)
NB_BIDDERS		-0.004 (0.007)		-0.061** (0.027)		-0.092 (0.084)
OPTION_DUM		0.098** (0.040)		-0.252 (0.158)		-0.884** (0.355)
BILL_BOND_DUM		0.097*** (0.019)		-0.108 (0.069)		1.006*** (0.194)
MIN_TRADING_LOT		0.000 (0.001)		-0.006*** (0.002)		-0.013** (0.006)
LEND_FAC		-0.024*** (0.007)		-0.001 (0.020)		0.086 (0.096)
COMISSION		0.050** (0.025)		0.109 (0.145)		-0.646 (0.613)
TREA_BOND_VOL		0.000 (0.001)		0.004 (0.004)		-0.039** (0.017)
5Y_SPREAD		-0.013 (0.012)		0.002 (0.029)		0.177 (0.115)
CONTROL_VAR	No	Yes	No	Yes	No	Yes
Bidder ID dummy	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	2,083	1,945	1,104	1,085	2,088	1,950
Adjusted R-squared	2.0	7.1	5.5	17.2	3.1	10.1

Note: This table presents the results from regressing three bidder-level variables on the auction method and a set of explanatory variables. We calculate bid dispersion, aggressive bid fraction, and bid premium for each bidder. The first dependent variable is bid dispersion. It is the quantity-weighted standard deviation of the bids submitted by the bidder. The second dependent variable is the fraction of aggressive bids. A bid is categorized as aggressive if it exceeds the closing price recorded in the secondary market before the auction. The third dependent variable is the bid premium. It is defined as the quantity-weighted average bid minus the closing price recorded in the secondary market before the auction. The regressions in columns 2, 4, and 6 include the auction method and the bidder ID dummies. All control variables are included in columns 3, 5, and 7. Appendix 2 provides the explanatory variable definitions. Standard errors, reported below the coefficients, are adjusted for data heteroscedasticity with the Huber-White correction. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

In this subsection, we verify if treasury security prices evolve differently around auction days under the discriminatory and uniform-price methods. We calculate $P_i(t)/P_i(0) - 1$ on each day t surrounding the auction i , with $t \in [-5, 5]$, and $t = 0$ is the auction day. Figure 3 shows the average impact $1/n \times \sum_{i=1}^n [P_i(t)/P_i(0) - 1]$ under the two auction methods. Given the necessity of observing a secondary price before the auction, the sample used to

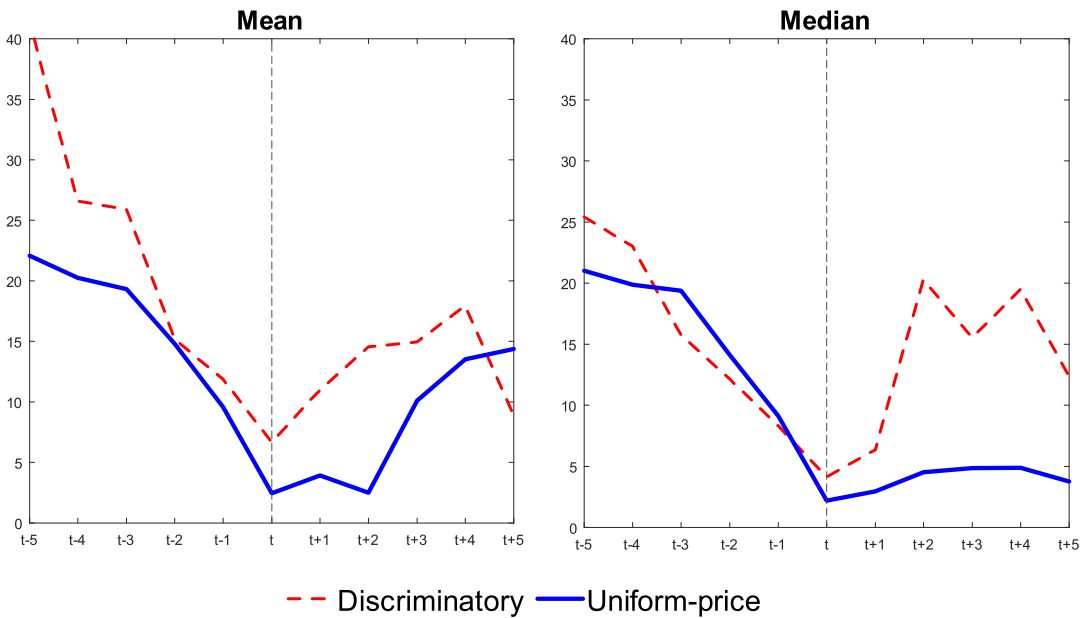


FIGURE 3 Secondary market price around treasury bond issuances. This figure illustrates the evolution of the secondary market price relative to the auction outcome prices surrounding a treasury bond issuance. The auction outcome price is the quantity-weighted average winning price (stop-out price) under the discriminatory (uniform-price) method. For five days before and after the auction, we compute $P_i(t)/P_i(0) - 1$, where $P_i(t)$ is the treasury bond price on day t , and $t = 0$ is the auction day i . We track the same bonds before and after the auctions. The sample includes reopening auctions only and excludes the Icelandic financial crisis period. The solid (dotted) line corresponds to the auctions made under the uniform-price (discriminatory) method. X-axis: The timeline around the issuance day t . Y-axis: $P_i(t)/P_i(0) - 1$ expressed in basis points. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jfr.12428)]

generate the figure contains reopening auctions only. The solid (dashed) corresponds to the set of auctions organized under the uniform-price (discriminatory) method. For both methods, we observe the same pattern. First, there is a sharp decline in the secondary market price as we move closer to the auction day, followed by a gradual recovery after the auction. It produces a V-shaped pattern around the auction day. This pattern is consistent with what Lou, et al. (2013) and Amin and Tédongap (2023) find in the context of US treasury auctions, and Beetsma, Giuliodori, de Jong, and Widijanto (2016) and Sigaux (2024) for European treasury auctions.¹⁶ Beetsma, et al. (2016) calls this mechanism the “auction cycle”. Before the auction, some market participants open short positions on forward contracts for soon-to-deliver treasury securities. They expect to make delivery with the treasury securities acquired on the auction. The unusually high open short positions produce downward price pressure before the auction. The price pressure is temporary and the price reverts to its pre-auction levels.

To assess the significance of the price impact surrounding the treasury auctions, we calculate the t-statistics associated with $P_i(t)/P_i(0) - 1$ on each day $t = 1, \dots, 5$ following the auctions. We exclude the auctions organized under the Icelandic financial crisis to avoid the leverage effect they may produce on prices. Table 9 reports average price impacts, t-stats, and significance levels. We find significant price impacts for discriminatory auctions only. The absence of significance for the observations under the uniform-price method shows that the arrival of this method has led to an impressive reduction of the price impact, up to the point that it is no longer statistically different from zero. We conclude that, under the uniform-price method, the dramatic reduction of the price impact in the secondary market combined with more aggressive bids in the primary market (previous subsection analysis) jointly generates lower underpricing for this method.

¹⁶This is a standard (inverted) V-shaped when price impacts are calculated in price (yield) space.

TABLE 9 Price impact of a treasury issuance.

t	Discriminatory Method		Uniform-Price Method	
1	0.11%	3.00***	0.02%	0.89
2	0.14%	2.11**	0.02%	0.48
3	0.15%	1.65	0.09%	1.3
4	0.18%	1.77*	0.12%	1.51
5	0.09%	0.61	0.13%	1.46

Note: This table reports the average price impact that follows a treasury bond issuance. A price impact is computed as $P_i(t)/P_i(0) - 1$, where $P_i(t)$ is the treasury bond price observed in the secondary market on day t , and $t = 0$ is the auction day i . We track the same treasury bonds before and after the auctions. The sample includes reopening auctions only. T-stats are reported in columns 3 and 5. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

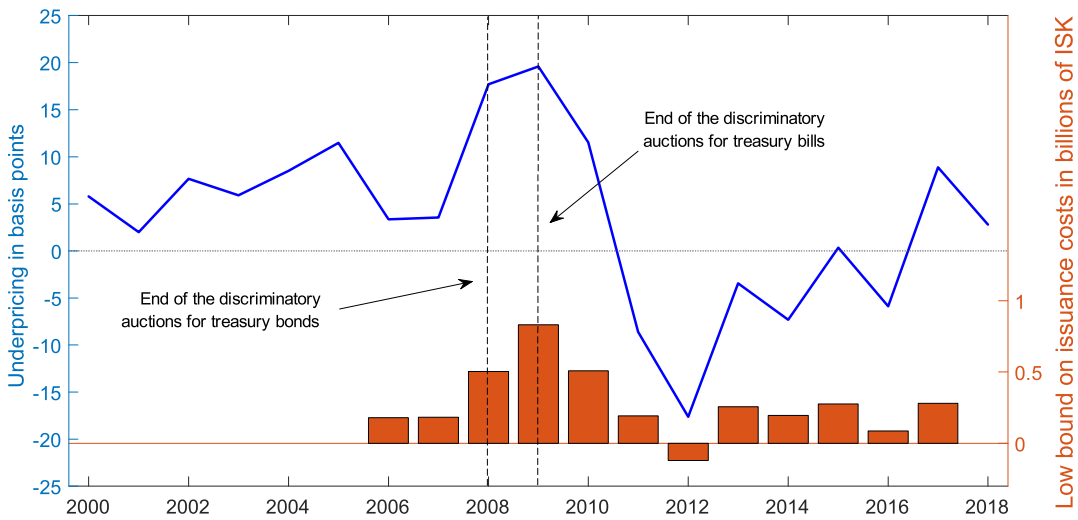


FIGURE 4 Underpricing and issuance costs in the Icelandic treasury security market. This graph reports the annual average level of underpricing for the Icelandic Treasury securities (left axis) and the corresponding estimation of the issuance cost incurred by the Icelandic Government (right axis). The first T-bill (T-bond) auction under the uniform-price method took place the 10/13/2009 (12/3/2008). We define underpricing as the auction outcome price minus the secondary market price. The auction outcome price is the quantity-weighted average winning price under the discriminatory method and the stop-out price under the uniform-price method. If there is no closing price in the secondary market on the auction day, we use the closest available closing price within the next five days instead. For each auction, the issuance cost is measured as underpricing times the auction supply plus the commissions paid to the PDs in the auction (presented in Subsection 3.1). The annual issuance cost is calculated as the sum of the issuance costs across auctions plus the annual commissions paid to PDs for their market-making activity in the secondary market. Our measure is an estimation because we cannot calculate underpricing for all auctions. For 2000–2005, the absence of clear information on commissions makes it not possible to estimate the issuance costs. For 2018, our dataset contains auction data up to March only. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

6.3 | Underpricing dynamics, commissions, and issuance costs

Figure 4 shows the annual average underpricing (line) from 2000 to 2017. What catches our attention is the emergence of negative underpricing, referred to as overpricing, for 2011–2014 and 2016. For 2014, it even concerns more than 50% of the auctions. Although this phenomenon is not limited to Iceland, it is surprising to

see overpricing.¹⁷ It suggests a negative income for the PDs/market-makers because the price paid to acquire the security in the auction is above what PDs can pocket by reselling in the secondary market. Given this, why would PDs participate in auctions if they lose money? We posit that the emergence of overpricing is linked to the existence of the remuneration plan. If the auctioneer pays commissions, overpricing can exist and be sustained as long as these commissions are large enough to guarantee a positive net revenue from participating in the treasury market. Our claim finds roots in Ding, Xiong, and Zhang (2022), who bring evidence that overpricing dropped following the Chinese government's decision to prohibit underwriters from using rebates in issuances of corporate debt securities in October 2017. Pacini (2009) proposes to link overpricing to the rights and privileges granted to primary dealers by treasury agencies, a hypothesis called “commodity bundling.”

The underpricing level is the usual metric that proxies the issuance costs incurred by the governments. We propose combining commissions and underpricing to generate a total issuance cost. If our claim is correct, this total issuance cost should remain positive over the sample period. We start by computing the issuance cost at the auction level. The first amount equals the secondary market price minus the auction outcome price times the number of securities issued. The second amount is the commissions paid to the PDs on their winning bids. Then, we obtain an annual amount by summing all the auction-level issuance costs. We add the commissions paid to the PDs for their market-making activity to this number. Although our measure is an approximation,¹⁸ our paper is, to the best of our knowledge, the first to calculate issuance costs by taking commissions explicitly into account. In Figure 4, the bars show the annual issuance costs from 2006 to 2018.¹⁹ The former represents an average of 281 ISK million, or 0.26%, by treasury securities issued. If we measure issuance costs without commissions, this number falls to 0.04%. The respective contributions of underpricing and commissions to the annual issuance cost are 34% and 66% on average. Except for 2012, the annual issuance costs are consistently above zero when overpricing occurs. Hence, the total payment in terms of remuneration exceeds the amount of money that represents overpricing.

Future research could apply our approach to all other countries offering a remuneration plan for their primary dealers. As far as we know, the other treasury agencies stating on their website the existence of a remuneration plan for their PDs are those of the following countries: Hungary, Italy, Norway, and Sweden. However, many other countries pay either commission (within auctions) or fees (within syndicated offerings) and do not communicate about it. While numerous studies have solely focused on underpricing to infer conclusions on the issuance cost, our finding shows that the rich content of the overall sovereign debt issuance design is essential. First, future studies should incorporate the expenses that represent the commissions paid to PDs. Beyond this, the indirect costs for the issuer that represent the rights and privileges granted to the PDs should also be addressed, as Pacini (2009) suggests.

7 | CONCLUSION

In 2008, the Central Bank of Iceland gave up the discriminatory method and adopted the uniform-price method to auction treasury securities. Using a dataset containing 516 auctions organized between 2000 and 2018, we assess whether adopting the uniform-price method changes the underpricing level in a significant way. Although this research question has been addressed before, our study is unique in three ways: (i) unlike previous studies with limited sample periods, our sample spans almost two decades with detailed auction and secondary market data, (ii) this study is the first to focus on a treasury security market with few primary dealers, a feature shared by many emerging markets but often overlooked in US-focused studies, (iii) we incorporate a rich set of control variables in the analysis and identify the first-order effect of auction-related factors in explaining underpricing.

¹⁷Pacini (2009) and Álvarez and Mazón (2019) have found overpricing for Italian and Spanish treasury auctions, respectively.

¹⁸There are no secondary market prices for a few securities issued.

¹⁹We did not find any archives about the PD remuneration plan before 2006.

We find that the Icelandic government benefits from adopting the uniform-price method. This result is statistically significant in multivariate regressions. It is economically important, too: the underpricing decline represents millions of savings on the cost of debt. Our study, taken jointly with similar analyses done by Goldreich (2007) and Umlauf (1993), supports the superiority of the uniform method when the variable of interest is underpricing. Does our result mean that the uniform-price method reduces the total issuance costs incurred by a government? Not necessarily, but issuance costs are broader than auction underpricing and can include explicit commissions, syndication fees, and other privileges awarded to primary dealers. A comprehensive estimation depends on the rules that frame or govern the relationships between the auctioneer and the primary dealers. For the Icelandic case, the growing commissions paid to the primary dealers between 2009 and 2018 offset the monetary gain induced by lower auction underpricing. While our results support the uniform-price method in reducing auction underpricing, we call for future research to consider total issuance costs and to investigate the richness of sovereign debt issuance's institutional settings.

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APPENDIX 1: GOVERNMENT DEBT MANAGEMENT DATA

The website www.lanamal.is provides current and historical documents on the terms and conditions of the issuance process, the secondary market rule, and primary dealer signing agreements. Minimum trading lots (Table A1.1) and commissions (Table A1.2) from 2000 to 2018 come from the documents "Agreement Concerning Issuance of Treasury Securities and Market Making in the Secondary Market". Two clarifications about commissions are essential. First, the documents report the annual aggregate amount paid to all primary dealers, not the exact amount paid to each primary dealer. Second, this amount is the only source of remuneration. There are no syndicated offerings on the Icelandic Treasury market and hence no syndication fees. Lending facility data (Table A1.3) comes

TABLE A1.1 Minimum trading lots.

Beginning date	Ending date	Securities	Minimum trading lot
6/1/2000	5/31/2001	Bonds	30
6/1/2000	5/31/2001	Bills	100
6/1/2001	5/31/2002	Bonds	50
6/1/2002	5/31/2003	Bonds	60
7/1/2003	5/31/2004	Bills	100
7/1/2003	5/31/2004	Bonds	60
6/1/2004	5/31/2005	Bills and bonds	100
5/30/2006	5/31/2007	Bills and bonds	100
6/1/2007	5/31/2008	Bills and bonds	100
12/1/2008	5/31/2009	Bills and bonds	50
6/1/2009	3/31/2018	Bills and bonds	100

Note: This table reports the minimum trading lot the primary dealers are obliged to propose in millions of Icelandic Krona before the trading day opens. We collect the data from the files "Agreement Concerning Issuance of Treasury Securities and Market Making in the Secondary Market".

TABLE A1.2 Commissions paid to primary dealers.

Beginning date	Ending date	Security	Commissions
6/1/2001	5/31/2002	Bonds	105
6/1/2002	5/31/2003	Bonds	112
6/1/2003	5/31/2004	Bills	55
6/1/2003	5/31/2004	Bills	112
6/29/2004	5/29/2006	Bills	52
6/1/2004	5/29/2006	Bonds	102
5/30/2006	5/31/2007	Bonds	160
6/1/2007	5/31/2008	Bonds	160
12/1/2008	5/31/2009	Bonds	160
6/1/2009	5/31/2010	Bonds	100

TABLE A1.2 (Continued)

Beginning date	Ending date	Security	Commissions
6/1/2010	5/31/2011	Bonds	110
6/1/2011	3/31/2012	Bonds	150
4/1/2012	3/31/2013	Bonds	180
4/1/2013	3/31/2014	Bonds	180
4/1/2014	3/31/2015	Bonds	180
4/1/2015	3/31/2016	Bonds	180
4/1/2016	3/31/2017	Bonds	180
4/1/2017	3/31/2018	Bonds	180

Note: This table reports the commissions paid to the primary dealers for their market-making activity in millions of Icelandic Krona. We collect the data from the files "Agreement Concerning Issuance of Treasury Securities and Market Making in the Secondary Market" disclosed by the CBI. The beginning (ending) dates refer to when the agreement takes effect (expires). The numbers reported are the total amounts at all primary dealers' disposal. The fraction earned by a primary dealer depends on his market share among all the transactions recorded on the trading system during the first six months of the agreement.

TABLE A1.3 Maximum loan granted to primary dealers.

Treasury bonds	Start	End	Loan	Treasury bonds	Start	End	Loan
RIKB 07 0209	1/1/2001	3/19/2003	0.4	RIKB 10 1210	9/26/2008	6/2/2009	5
	3/20/2003	8/31/2004	1		6/3/2009	4/5/2010	3
	9/1/2004	6/19/2005	1		4/6/2010	5/13/2010	3
	6/20/2005	5/30/2006	1.2		5/14/2010	5/31/2010	5
	5/30/2006	2/9/2007	2		6/1/2010	12/10/2010	3
RIKB 08 0613	5/30/2006	1/4/2007	2	RIKB 11 0722	6/3/2009	4/5/2010	3
	1/5/2007	9/5/2007	3		4/6/2010	5/31/2010	3
	9/6/2007	1/30/2008	3		6/1/2010	12/9/2010	2
	1/31/2008	6/13/2008	5		12/10/2010	7/22/2011	5
RIKB 08 1212	5/30/2006	1/4/2007	2	RIKB 12 0824	6/1/2010	4/28/2011	1*
	1/5/2007	9/5/2007	5		4/29/2011	7/21/2011	2
	9/6/2007	1/30/2008	3		7/22/2011	8/24/2012	5
	1/31/2008	5/12/2008	5	RIKB 13 0517	9/1/2004	6/19/2005	1
	5/13/2008	9/25/2008	7		6/20/2005	5/30/2006	1.2
	9/26/2008	12/12/2009	7		5/30/2006	1/4/2007	2
RIKB 09 0612	1/5/2007	9/5/2007	3		1/5/2007	9/5/2007	3
	9/6/2007	1/30/2008	3		9/6/2007	1/30/2008	3
	1/31/2008	5/12/2008	5		1/31/2008	9/25/2008	5

(Continues)

TABLE A1.3 (Continued)

Treasury bonds	Start	End	Loan	Treasury bonds	Start	End	Loan
	5/13/2008	9/25/2008	7		9/26/2008	6/2/2009	5
	9/26/2008	10/7/2008	7		6/3/2009	4/5/2010	3
	10/8/2008	6/2/2009	8		4/6/2010	5/31/2010	3
	6/3/2009	6/12/2009	3		6/1/2010	3/15/2012	3
					3/16/2012	8/23/2012	3
RIKB 10 0317	9/1/2004	6/19/2005	1		8/24/2012	5/17/2013	5
	6/20/2005	5/30/2006	1.2				
	5/30/2006	1/4/2007	2	RIKB 14 0314	3/9/2012	10/9/2012	1
	1/5/2007	9/5/2007	3		10/10/2012	3/14/2013	2
	9/6/2007	1/30/2008	3		3/15/2013	5/15/2013	2
	1/31/2008	9/25/2008	5		5/16/2013	3/14/2014	4
	9/26/2008	10/7/2008	5				
	10/8/2008	6/2/2009	8	RIKB 15 0408	3/15/2013	7/9/2013	1
	6/3/2009	3/17/2010	3		7/10/2013	3/20/2015	2
					3/31/2015	4/8/2015	4
Treasury bonds	Start	End	Loan				
RIKB 16 1013	6/1/2010	3/20/2015	1*				
	3/31/2015	10/13/2016	1*				
RIKB 17 0206	2/6/2015	11/24/2015	1				
	11/25/2015	2/6/2017	2				
RIKB 19 0226	1/31/2008	9/25/2008	5				
	9/26/2008	6/2/2009	5				
	6/3/2009	4/5/2010	3				
	4/6/2010	5/31/2010	3				
	6/1/2010	3/20/2015	1*				
	3/31/2015	6/25/2017	1*				
	6/26/2017	3/31/2018	1*				
RIKB 20 0205	6/1/2010	3/11/2014	1				
	3/12/2014	9/9/2014	1				
	9/10/2014	3/30/2015	2				
	3/31/2015	3/31/2018	1*				
RIKB 22 1026	6/1/2010	3/20/2015	1*				
	3/31/2015	6/25/2017	1*				
	6/26/2017	3/31/2018	1*				
RIKB 25 0612	6/3/2009	4/5/2010	3				

TABLE A1.3 (Continued)

Treasury bonds	Start	End	Loan
	4/6/2010	5/31/2010	3
	6/1/2010	3/20/2015	1*
	3/31/2015	6/25/2017	1*
	6/26/2017	3/31/2018	1*
RIKB 28 1115	3/31/2015	1/25/2017	1
	1/26/2017	3/7/2017	1
	3/8/2017	3/31/2018	2
RIKB 31 0124	6/1/2010	3/20/2015	1*
	3/31/2015	6/25/2017	1*
	6/26/2017	3/30/2018	1*
Treasury bills	Start	End	Loan
All	1/1/2000	1/4/2006	0
	1/5/2006	1/4/2007	1
	1/5/2007	9/5/2007	3
	9/6/2007	1/30/2008	0
	1/31/2008	9/25/2008	5
	9/26/2008	6/2/2009	5
	6/3/2009	4/5/2010	3
	4/6/2010	3/31/2018	0
RIKV 07 0301	9/6/2007	1/30/2008	3
RIKV 07 0402	9/6/2007	1/30/2008	3

Note: This table reports the maximum amount, in billions of Icelandic Krona, loaned by the CBI in each treasury security for each primary dealer. We collect the data manually from the files "Agreement Concerning Issuance of Treasury Securities and Market Making in the Secondary Market". The beginning (ending) dates refer to when the agreement takes effect (expires). If the security matures before the agreement ends, the maturity date replaces the agreement expiration date. *: In this series, the maximum amount loaned to each primary dealer is one billion ISK nominal value but rises to two billion when the issue has reached 20 billion.

from the documents "Terms and Conditions for Securities Lending Facilities for Primary Dealers". These documents indicate the maximum amount each primary dealer can borrow for each treasury security. This amount varies with the security outstanding amount and is therefore updated regularly. All marketable treasury securities are eligible for securities loans.

APPENDIX 2: EXPLANATORY VARIABLE DEFINITIONS

AUCTION_METHOD

An indicator variable equal to one (zero) for the auctions organized under the uniform-price (discriminatory) method. The last T-bill (T-bond) auction under the discriminatory method occurred on 9/11/2009 (9/25/2008). The first T-bill (T-bond) auction under the uniform-price method occurred on 10/13/2009 (12/3/2008).

AUCTION_SIZE

The natural logarithm of the number of treasury securities issued in the auction.

BID_TO_COVER

The demand for treasury securities formulated by the primary dealers in the auction divided by the number of securities issued.

BILL_BOND_DUM

An indicator variable equal to one if the issued security is a treasury bond, and zero otherwise.

COMMISSION

The commissions paid to the primary dealers for their market-making activity in the secondary market, in logarithm. Data are reported in Table [A1.2](#).

CONTROL_VAR

A set of two variables: ICEX_RTN and ICEX_VOL.

BID_DISPERSION

The bid dispersion, calculated at the auction level. First, we compute the quantity-weighted standard deviation of the bids for each primary dealer. Then, we sum the squared values of the primary dealer bid dispersions. The auction-level dispersion is calculated as the square root of this sum.

ICEX_RTN

The 60-day stock index return computed with the Iceland All Shares PI index.

ICEX_VOL

The 60-day index volatility computed with the Iceland All Shares PI index.

LEND_FAC

The maximum loan amount by treasury security and by the primary dealer. Data are reported in Table [A1.3](#).

MIN_TRADING_LOT

The minimum trading lot that each primary dealer is obliged to propose in the secondary market within their market-making activity. Data are reported in Table [A1.1](#).

AWARD_CONCEN

The number of treasury securities awarded to the largest winner in the auction as a percentage of the number of securities issued.

NB_BIDDERS

The number of bidders in the auction.

REOP_CYCLE_DUM

An indicator variable equal to one if the treasury security has already been auctioned before, and zero otherwise.

OPTION_DUM

An indicator variable equal to one after June 2007, and zero otherwise.

TREA_BOND_VOL

The monthly volatility of the Icelandic treasury bond market one month prior to the auction date. It is calculated with the prices of the issue RIKB 130517 from 5/21/2002 to 1/26/2011 and RIKB 310124 from 1/27/2011 to 3/28/2018.

5Y_SPREAD

The difference between the Icelandic and German 5-year Treasury bond yield. The data are from Capital IQ.